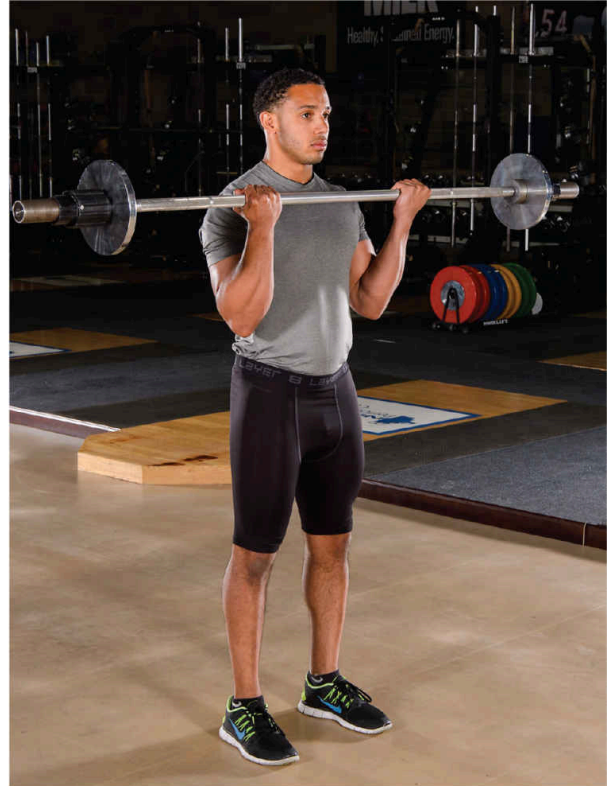




Starting position



Upward and downward movements

11

HAMMER CURL

Starting Position

- Grasp two dumbbells using a closed, neutral grip.
- Stand erect with the feet shoulder-width apart and the knees slightly flexed.
- Position the dumbbells alongside the thighs with the elbows fully extended.
- All repetitions begin from this position.

Upward Movement Phase

- Keeping the dumbbell in a neutral grip, flex the elbow of one arm until the dumbbell is near the anterior deltoid. The other arm should be kept stationary at the side of the thigh.
- Keep the torso erect and the upper arm stationary.
- Do not jerk the body or swing the dumbbell upward.

Downward Movement Phase

- Lower the dumbbell until the elbow is fully extended.
- Keep the dumbbell in a neutral grip position.
- Keep the torso and knees in the same position.
- Repeat the upward and downward movement phases with the other arm (alternate arms).

MAJOR MUSCLES INVOLVED

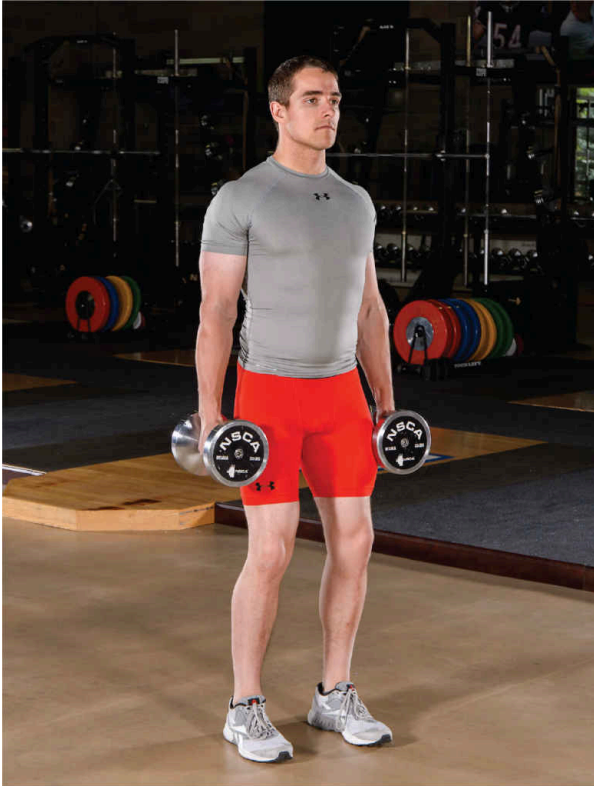
brachialis, biceps brachii, brachioradialis

VERBAL CUES

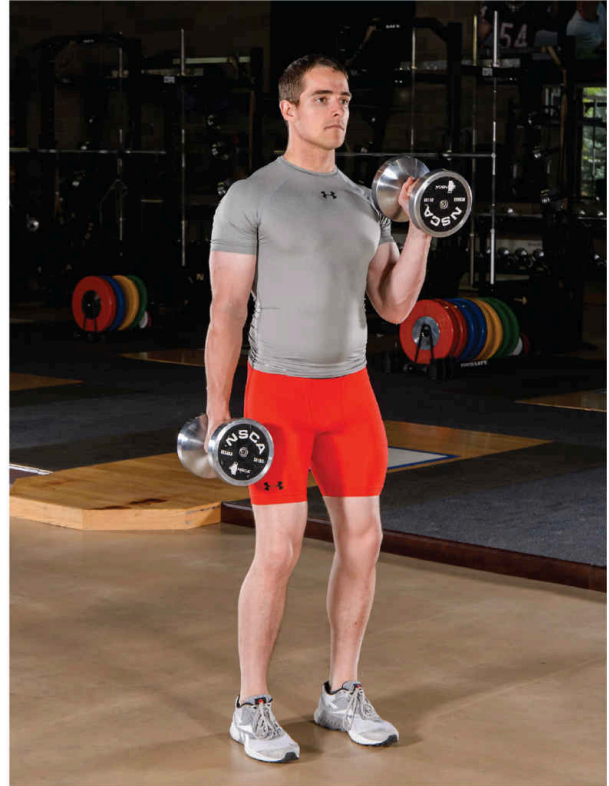
- **Curl up, thumbs to shoulders**
- **Control the descent; imagine crushing something between your biceps and forearm**

COMMON ERRORS

- **Swinging the body or using a rocking motion instead of a controlled motion**
- **Not using a full ROM by not lowering the weight fully or not curling the weight high enough**



Starting position



Upward and downward movements

Calves

12 STANDING CALF (HEEL) RAISE (MACHINE)

Starting Position

- Position the body evenly under the shoulder pads.
- Grasp the handles and place the balls of the feet on the nearest edge of the step with the legs and feet hip-width apart and parallel to each other.
- Stand erect with the knees fully extended but not forcefully locked out.
- Allow the heels to lower to a comfortable, stretched position.
- All repetitions begin from this position.

Upward Movement Phase

- Keeping the torso erect and the legs and feet parallel, push up as high as possible on the toes.
- Push up off the step; do not invert or evert the feet.
- Keep the knees extended but not locked out.

Downward Movement Phase

- Allow the heels to lower slowly back to the starting position.
- Maintain the same body position.

MAJOR MUSCLES INVOLVED

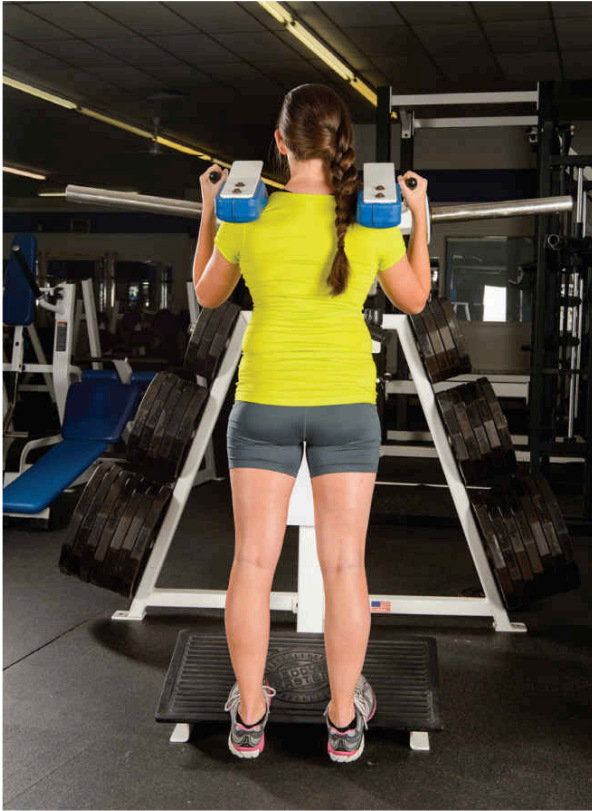
gastrocnemius, soleus

VERBAL CUES

- **Push through your toes**
- **Reach for the ceiling**
- **Control the descent**

COMMON ERRORS

- **Insufficient ROM**
- **Significant knee flexion**



Starting position



Upward and downward movements

13 SEATED CALF (HEEL) RAISE (MACHINE)

Starting Position

- Sit erect on the seat and place the balls of the feet on the nearest edge of the step with the legs and feet hip-width apart and parallel to each other.
- Lower the thigh-knee pad so it firmly presses against the knees and front of the lower thigh area.
- Plantar flex the ankles to remove the supports.
- Allow the heels to lower to a comfortable, stretched position.
- All repetitions begin from this position.

Upward Movement Phase

- Keeping the torso erect and the legs and feet parallel, push up on the toes as high as possible.
- Push up off the step; do not invert or evert the feet.

Downward Movement Phase

- Allow the heels to lower slowly back to the starting position.
- Maintain the same body position.
- At the end of the set, return the supports to the starting position and remove the feet.

MAJOR MUSCLES INVOLVED

soleus, gastrocnemius

VERBAL CUES

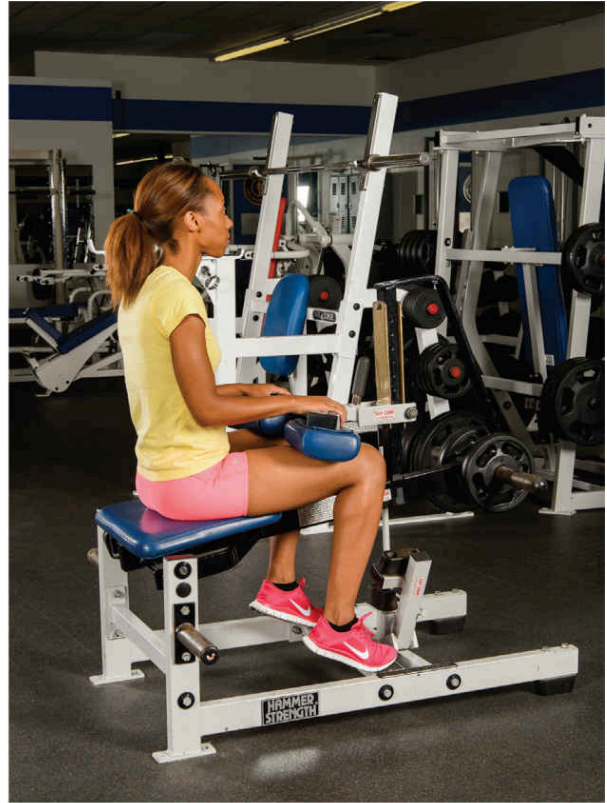
- **Push through your toes**
- **Squeeze at the top**
- **Control the descent**

COMMON ERRORS

- **Pulling with the hands or jerking the torso to help raise the weight**
- **Bouncing at the bottom of the movement instead of performing controlled repetitions**



Starting position



Upward and downward movements

Chest

14 FLAT BARBELL BENCH PRESS (AND DUMBBELL VARIATION)

This exercise can also be performed with two dumbbells, using a closed, pronated grip. When using dumbbells, the spotter assists by spotting the athlete's forearms near the wrists instead of the bar.

Starting Position: Athlete

- Lie in a supine position on a bench in the five-point body contact position.
- Place the body on the bench so that the eyes are below the racked bar.
- Grasp the bar with a closed, pronated grip slightly wider than shoulder-width apart.

- Signal the spotter for assistance in moving the bar off the supports.
- Position the bar over the chest with the elbows fully extended.
- All repetitions begin from this position.

Starting Position: Spotter

- Stand erect and very close to the head of the bench (but do not distract the athlete).
- Place the feet shoulder-width apart with the knees slightly flexed.
- Grasp the bar with a closed, alternated grip inside the athlete's hands.
- At the athlete's signal, assist with moving the bar off the supports.
- Guide the bar to a position over the athlete's chest.
- Release the bar smoothly.

Downward Movement Phase: Athlete

- Lower the bar to touch the chest at approximately nipple level.
- Keep the wrists stiff and the forearms perpendicular to the floor and parallel to each other.
- Maintain the five-point body contact position.

Downward Movement Phase: Spotter

- Keep the hands in the alternated grip position close to—but not touching—the bar as it descends.
- Slightly flex the knees, hips, and torso and keep the back neutral when following the bar.

Upward Movement Phase: Athlete

- Push the bar upward and very slightly backward until the elbows are fully extended.
- Keep the wrists stiff and the forearms perpendicular to the floor and parallel to each other.
- Maintain the five-point body contact position.

- Do not arch the back or raise the chest to meet the bar.
- At the end of the set, signal the spotter for assistance in racking the bar.
- Keep a grip on the bar until it is racked.

Upward Movement Phase: Spotter

- Keep the hands in the alternated grip position close to—but not touching—the bar as it ascends.
- Slightly extend the knees, hips, and torso and keep the back neutral when following the bar.
- At the athlete's signal, grasp the bar with an alternated grip inside the athlete's hands.
- Guide the bar back onto the supports.
- Keep a grip on the bar until it is racked.

MAJOR MUSCLES INVOLVED

pectoralis major, anterior deltoids, triceps brachii

VERBAL CUES

- **Push yourself away from the bar**
- **Try to bend the bar; touch and go**

COMMON ERRORS

- **Bouncing the bar off the chest during the upward movement phase**
- **Lifting the hips off the bench**
- **Not wrapping the thumbs around the barbell**



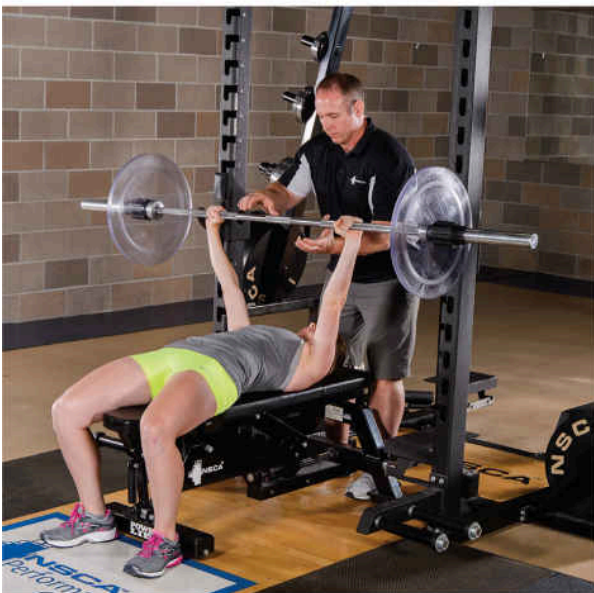
Liftoff



Starting positions



Downward movements



Upward movements



Racking the bar

15 INCLINE DUMBBELL BENCH PRESS (AND BARBELL VARIATION)

This exercise can also be performed with a barbell, using a closed, pronated grip slightly wider than shoulder width. When using a barbell, the spotter assists by spotting the bar instead of the forearms near the athlete's wrists.

Starting Position: Athlete

- Grasp two dumbbells using a closed, pronated grip.
- Lie in a supine position on an incline bench in the five-point body contact position.
- Signal the spotter for assistance in moving the dumbbells into the starting position.
- Press the dumbbells in unison to an extended-elbow, parallel-arm position above the head and face.
- All repetitions begin from this position.

Starting Position: Spotter

- Stand erect and very close to the head of the bench (but do not distract the athlete).
- Place the feet shoulder-width apart with the knees slightly flexed.
- Grasp the athlete's forearms near the wrists.
- At the athlete's signal, assist with moving the dumbbells to a position over the athlete's head and face.
- Release the athlete's forearms smoothly.

Downward Movement Phase: Athlete

- Lower the dumbbells down and slightly out to be near the armpits and in line with the upper one-third area of the chest (between the clavicles and the nipples).
- Keep the wrists stiff and directly above the elbows, with the dumbbell handles aligned with each other.
- Maintain the five-point body contact position.
- Do not arch the back or raise the chest to meet the dumbbells.

Downward Movement Phase: Spotter

- Keep the hands near—but not touching—the athlete's forearms near the wrists as the dumbbells descend.
- Slightly flex the knees, hips, and torso and keep the back neutral when following the dumbbells.

Upward Movement Phase: Athlete

- Push the dumbbells upward at the same rate and very slightly toward each other until the elbows are fully extended.
- Keep the wrists stiff and directly above the elbows, with the dumbbell handles aligned with each other.
- Maintain the five-point body contact position.

Upward Movement Phase: Spotter

- Keep the hands near—but not touching—the athlete's forearms near the wrists as the dumbbells ascend.
- Slightly extend the knees, hips, and torso and keep the back neutral when following the dumbbells.

MAJOR MUSCLES INVOLVED

pectoralis major, anterior deltoids, triceps brachii

VERBAL CUES

- **Maintain a neutral wrist position**
- **Squeeze your chest at the top**
- **Control the descent**

COMMON ERRORS

- **Arching the lower back excessively**
- **Insufficient ROM**



Starting positions



Downward and upward movements

▶ **16** **FLAT DUMBBELL FLY (AND INCLINE VARIATION)**

This exercise can also be performed on an incline bench. If using the incline variation, begin by positioning the dumbbells over the head and face instead of over the chest.

Starting Position: Athlete

- Grasp two dumbbells using a closed, neutral grip.
- Lie in a supine position on a bench in the five-point body contact position.
- Signal the spotter for assistance in moving the dumbbells into the starting position.
- Press the dumbbells in unison to an extended-elbow position above the chest.
- Slightly flex the elbows and point them out to the sides.
- All repetitions begin from this position.

Starting Position: Spotter

- Position one knee on the floor with the foot of the other leg forward and flat on the floor (or kneel on both knees).
- Grasp the athlete's forearms near the wrists.
- At the athlete's signal, assist with moving the dumbbells to a position over the athlete's chest.
- Release the athlete's forearms smoothly.

Downward Movement Phase: Athlete

- Lower the dumbbells in a wide arc until they are level with the shoulders or chest.
- Keep the dumbbell handles parallel to each other as the elbows move downward.
- Keep the wrists stiff and the elbows held in a slightly flexed position.
- Keep the hands, wrists, forearms, elbows, upper arms, and shoulders in the same vertical plane.
- Maintain the five-point body contact position.

Downward Movement Phase: Spotter

- Keep the hands near—but not touching—the athlete's forearms near the wrists as the dumbbells descend.

Upward Movement Phase: Athlete

- Raise the dumbbells up toward each other in a wide arc back to the starting position.
- Keep the wrists stiff and the elbows held in a slightly flexed position.
- Keep the hands, wrists, forearms, elbows, upper arms, and shoulders in the same vertical plane.
- Maintain the five-point body contact position.

Upward Movement Phase: Spotter

- Keep the hands near—but not touching—the athlete's forearms near the wrists as the dumbbells ascend.

MAJOR MUSCLES INVOLVED

pectoralis major, anterior deltoids

VERBAL CUES

- **Open your arms like you are hugging a tree**
- **Maintain a slight elbow flex throughout the ROM**
- **Control the descent**

COMMON ERRORS

- **Keeping the elbows too extended**
- **Lowering the weights below the shoulder level**



Starting positions



Downward and upward movements



17

VERTICAL CHEST PRESS

(MACHINE)

Starting Position

- Sit down and lean back on the seat in the five-point body contact position.
- Grasp the handles with a closed, pronated grip.
- Align handles with the nipples. If necessary, adjust the seat height to correctly position the handles.
- All repetitions begin from this position.

Forward Movement Phase

- Push the handles away from the chest to a fully extended elbow position.
- Do not arch the lower back or forcefully lock out the elbows.
- Maintain the five-point body contact position.

Backward Movement Phase

- Allow the handles to slowly move backward to the starting position.
- Maintain the five-point body contact position.

MAJOR MUSCLES INVOLVED

pectoralis major, anterior deltoids, triceps brachii

VERBAL CUES

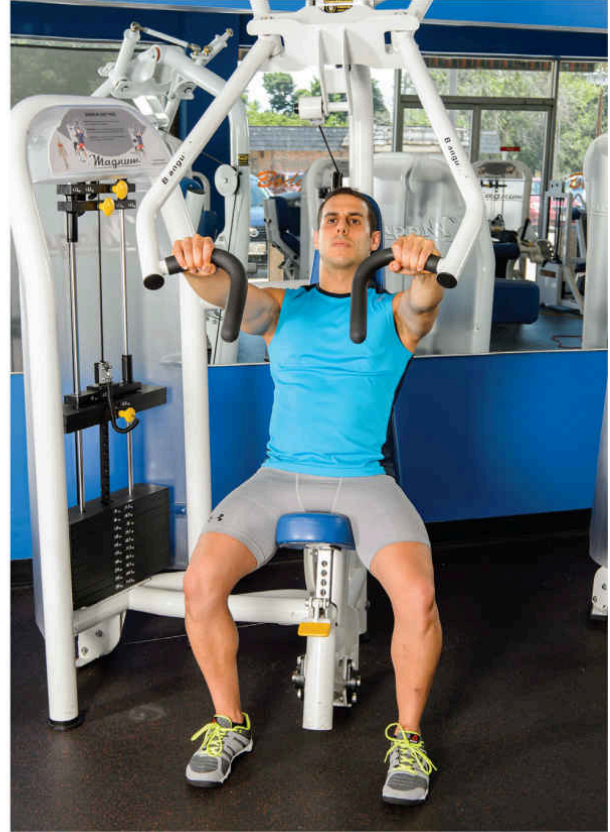
- **Back flat against the pad**
- **Feet flat on the floor**
- **Push the handles straight out**

COMMON ERRORS

- **Arching the back off the pad while performing the exercise**
- **Shortening the ROM during the backward movement phase**



Starting position



Forward and backward movements

18 PEC DECK (MACHINE)

Starting Position

- Sit down and lean back on the seat in the five-point body contact position. (*Note:* Some back pads are not long enough to allow the back of the head to be in contact with the bench.)
- Grasp the handles with a closed, neutral grip.
- Align the handles with the midchest so that the arms are parallel to the floor. If necessary, adjust the seat height to correctly position the handles.
- All repetitions begin from this position.

Forward Movement Phase

- Pull the handles toward each other while maintaining a slight flex in the elbow, until the fingers touch in front of the body.

- Do not arch the back or lock out the elbows.
- Maintain the five-point body contact position.

Backward Movement Phase

- Allow the handles to slowly move outward back to the starting position.
- Maintain the five-point body contact position.

MAJOR MUSCLES INVOLVED

pectoralis major, anterior deltoids, triceps brachii

VERBAL CUES

- **Back flat against the pad**
- **Squeeze your chest at the center**
- **Control the return**

COMMON ERRORS

- **Lifting the back off the pad, reducing chest engagement, and increasing lower back strain**
- **Incomplete ROM; not bringing handles together at the end of the movement**



Starting position



Forward and backward movements

Forearms

19 WRIST CURL

Starting Position

- Sit on the end of a bench.
- Grasp the bar with a closed, supinated grip about hip- to shoulder-width apart.
- Position the feet and legs parallel to each other with the toes pointing straight ahead.
- Lean the torso forward and place the elbows and forearms on the top of the thighs.
- Move the wrists forward until they extend slightly beyond the patellae.

- Allow the wrists to extend and the hands to open so that the fingertips hold the bar.

Upward Movement Phase

- Raise the bar by flexing the fingers and then the wrists.
- Flex the wrists as far as possible without moving the elbows or forearms.
- Do not jerk the shoulders backward or swing the bar upward.

Downward Movement Phase

- Allow the wrists and fingers to slowly extend back to the starting position.
- Keep the torso and arms in the same position.

MAJOR MUSCLES INVOLVED

flexor carpi ulnaris, flexor carpi radialis, palmaris longus

VERBAL CUES

- **Wrists off the edge of the bench**
- **Curl only from the wrists**
- **Full ROM**

COMMON ERRORS

- **Not lowering the wrist fully or not curling up completely**
- **Raising the forearms off the support surface during the movement**



Starting position



Upward and downward movements

20 WRIST EXTENSION

Starting Position

- Sit on the end of a bench.
- Grasp the bar with a closed, pronated grip about hip- to shoulder-width apart.
- Position the feet and legs parallel to each other with the toes pointing straight ahead.

- Lean the torso forward and place the elbows and forearms on the top of the thighs.
- Move the wrists forward until they are slightly beyond the patellae.
- Keep a closed grip on the bar but allow the wrists to flex toward the floor.

Upward Movement Phase

- Raise the bar by extending the wrists.
- Extend the wrists as far as possible without moving the elbows or forearms.
- Do not jerk the body backward or swing the bar upward.

Downward Movement Phase

- Allow the wrists to slowly flex back to the starting position.
- Keep the torso and arms in the same position.
- Maintain a closed grip.

MAJOR MUSCLES INVOLVED

extensor carpi ulnaris, extensor carpi radialis brevis (and longus)

VERBAL CUES

- **Keep your wrists aligned with your forearms**
- **Lift from the wrists only**
- **Curl your hands toward ceiling**

COMMON ERRORS

- **Insufficient forearm support and improper wrist alignment**
- **Allowing the wrist to rotate during the movement instead of maintaining a consistent plane**



Starting position



Upward and downward movements

Hips and Thighs



21

HIP SLED (MACHINE)

Starting Position

- Sit in the machine with the lower back, hips, and buttocks pressed against their pads.

- Place the feet flat on the platform hip-width apart with the toes slightly angled out.
- Position the legs parallel to each other.
- Grasp the handles or the sides of the seat and move the hips and knees to a fully extended position, but do not forcefully lock the knees.
- Keep the hips on the seat and the back pressed firmly and evenly against the back pad.
- Remove the support mechanism from the foot platform and grasp the handles or the seat again.
- All repetitions begin from this position.

Downward Movement Phase

- Allow the hips and knees to slowly flex to lower the platform.
- Do not allow the platform to lower rapidly.
- Keep the hips and buttocks on the seat and the back pressed firmly and evenly against the back pad.
- Keep the knees aligned over the feet as they flex.
- Allow the hips and knees to flex until the thighs are parallel to the foot platform.
- Do not allow the buttocks to lose contact with the seat, the hips to roll off the back pad, or the heels to rise off the foot platform.

Upward Movement Phase

- Push the platform up by extending the hips and knees.
- Push to a fully extended position, but do not forcefully lock out the knees.
- Maintain the same hip and back position; do not allow the buttocks to rise.
- Keep the knees aligned over the feet as they extend.
- At the end of the set, return the supports to the starting position, remove the feet, and exit the machine.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris

VERBAL CUES

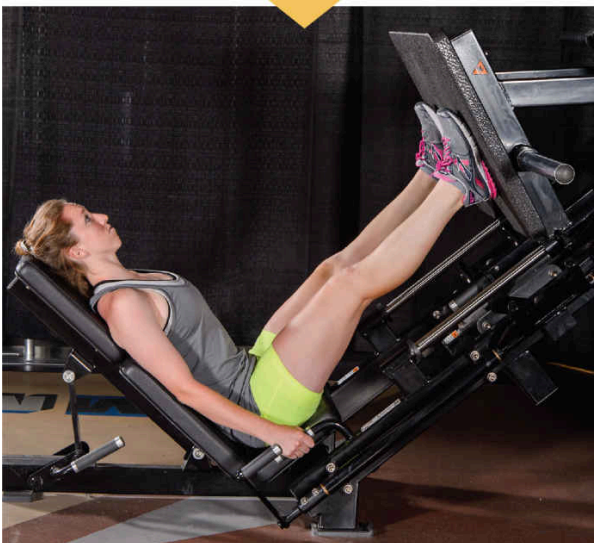
- **Toes pointed slightly outward with knees in line with the toes**
- **Lower the weight until the knees are at 90 degrees**
- **Push through the heel, driving the platform away**

COMMON ERRORS

- **Allowing the heels to lift off the platform**
- **Allowing the knees to move in (via hip adduction) or out (via hip abduction) during the movement**



Foot position



Starting position



Downward and upward movements

**22**

BACK SQUAT

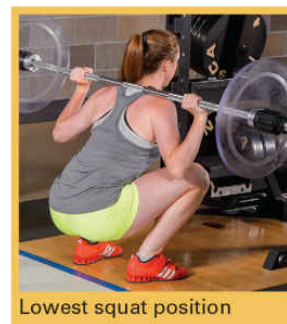
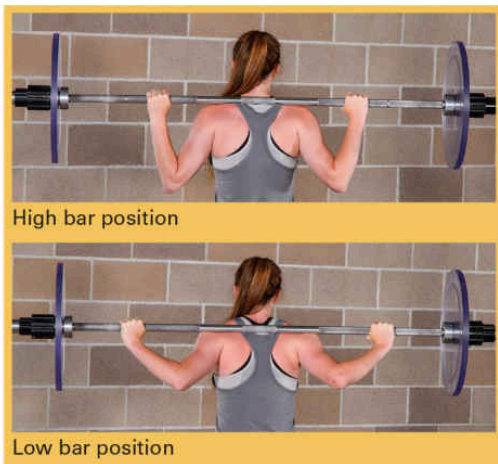
Starting Position: Athlete

- Grasp the bar with a closed, pronated grip (actual width depends on the bar position).
- Step under the bar and position the feet parallel to each other.
- Place the bar in a balanced position on the upper back and shoulders in one of two locations:
 - *Low bar position.* Across the posterior deltoids at the middle of the trapezius (using a handgrip wider than shoulder width)
 - *High bar position.* Above the posterior deltoids at the base of the neck (using a handgrip only slightly wider than shoulder width)
- Lift the elbows up to create a shelf for the bar using the upper back and shoulder muscles.
- Hold the chest up and out.
- Tilt the head slightly up.
- Once in position, signal the spotters for assistance in moving the bar off the supports.
- Extend the hips and knees to lift the bar.
- Take one or two steps backward.
- Position the feet shoulder-width apart (or wider), even with each other, with the toes pointed slightly outward.
- All repetitions begin from this position.

Starting Position: Two Spotters

- Stand erect at opposite ends of the bar with the feet shoulder-width apart and the knees slightly flexed.
- Grasp the end of the bar by wrapping the hands around the bar with the thumbs crossed and palms facing the barbell.
- At the athlete's signal, assist with lifting and balancing the bar as it is moved off the supports.

- Release the bar smoothly.
- Hold the hands 2 to 3 inches (5-8 cm) below the ends of the bar.
- Move sideways in unison with the athlete as the athlete moves backward.
- Once the athlete is in position, get into a shoulder-width stance with the knees slightly flexed and the torso erect.



Starting positions with high bar position



Downward movement positions

Downward Movement Phase: Athlete

- Maintain a position with the back neutral, elbows high, and the chest up and out.
- Allow the hips and knees to slowly flex while keeping the torso-to-floor angle relatively constant.
- Keep the heels on the floor and the knees aligned over the feet.
- Continue flexing the hips and knees until the tops of the thighs are parallel to the floor, the trunk begins to round or flex forward, or the heels rise off the floor.

Downward Movement Phase: Two Spotters

- Keep the thumbs crossed and hands close to—but not touching—the bar as it descends.
- Slightly flex the knees, hips, and torso to keep a neutral spine position when following the bar.

Upward Movement Phase: Athlete

- Maintain a position with neutral spine, high elbows, and the chest up and out.
- Extend the hips and knees at the same rate (to keep the torso-to-floor angle constant).
- Keep the heels on the floor and the knees aligned over the feet.
- Do not flex the torso forward or round the back.
- Continue extending the hips and knees to reach the starting position.
- At the end of the set, step forward toward the rack.
- Squat down until the bar rests on the supports.

Upward Movement Phase: Two Spotters

- Keep the thumbs crossed and hands close to—but not touching—the bar as it ascends.
- Slightly extend the knees, hips, and torso and keep the back neutral when following the bar.
- At the end of the set, move sideways in unison with the athlete back to the rack.

- Simultaneously grasp the bar and assist with balancing the bar as it is racked.
- Release the bar smoothly.

MAJOR MUSCLES INVOLVED

gluteus maximus, semimembranosus, semitendinosus, biceps femoris, vastus lateralis, vastus intermedius, vastus medialis, rectus femoris

VERBAL CUES

- **Chest up, head neutral**
- **Spread the floor with your feet**
- **Drive through your heels**

COMMON ERRORS

- **Allowing the heels to lift off the floor, the torso to flex further forward, or the upper back to round during the upward phase of the movement**
- **Allowing the knees to move in (via hip adduction) or out (via hip abduction) during the movement**



Upward movement positions



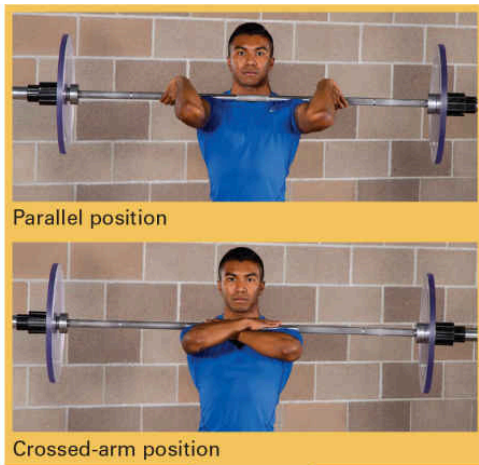
Racking the bar

23 FRONT SQUAT

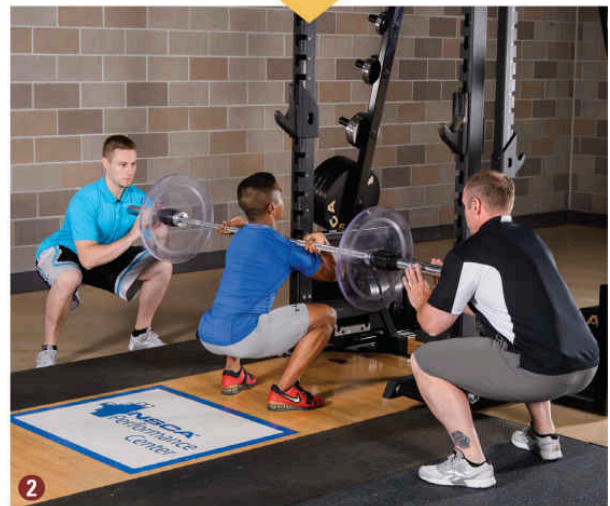
Starting Position: Athlete

- Step under the bar and position the feet parallel to each other.

- Place the hands on the bar in one of two arm positions:
 1. Parallel-arm position
 - » Grasp the bar with a closed, pronated grip.
 - » Grip should be slightly wider than shoulder width.
 - » Move up to the bar to place it on top of the anterior deltoids and clavicles.
 - » Fully flex the elbows to position the upper arms parallel to the floor.
 2. Crossed-arm position
 - » Flex the elbows and cross the arms in front of the chest.
 - » Move up to the bar to place it on top of the anterior deltoids.
 - » Use an open grip with the hands on top of the bar and the fingers holding it in place.
 - » Lift the elbows to position the arms parallel to the floor.
- Hold the chest up and out.
- Tilt the head slightly up.
- Once in position, signal the spotter for assistance in moving the bar off the supports.
- Extend the hips and knees to lift the bar.
- Take one or two steps backward.
- Position the feet shoulder-width apart (or wider), even with each other, with the toes pointed slightly outward.
- All repetitions begin from this position.



Starting positions with parallel arm position



Downward movement positions

Starting Position: Two Spotters

- Stand erect at opposite ends of the bar with the feet shoulder-width apart and the knees slightly flexed.
- Grasp the end of the bar by wrapping the hands around the bar with the thumbs crossed and palms facing the barbell.
- At the athlete's signal, assist with lifting and balancing the bar as it is moved off the supports.
- Release the bar smoothly.
- Hold the hands 2 to 3 inches (5-8 cm) below the ends of the bar.

- Move sideways in unison with the athlete as the athlete moves backward.
- Once the athlete is in position, get into a shoulder-width stance with the knees slightly flexed and the torso erect.

Downward Movement Phase: Athlete

- Maintain a position with the back neutral, elbows high, and the chest up and out.
- Allow the hips and knees to slowly flex while keeping the torso-to-floor angle relatively constant.
- Keep the heels on the floor and the knees aligned over the feet.
- Do not flex the torso forward or round the back.
- Continue flexing the hips and knees until the tops of the thighs are parallel to the floor, the trunk begins to round or flex forward, or the heels rise off the floor.

Downward Movement Phase: Two Spotters

- Keep the thumbs crossed and hands close to—but not touching—the bar as it descends.
- Slightly flex the knees, hips, and torso and keep the back neutral when following the bar.

Upward Movement Phase: Athlete

- Maintain a position with the back neutral, elbows high, and the chest up and out.
- Extend the hips and knees at the same rate (to keep the torso-to-floor angle constant).
- Keep the heels on the floor and the knees aligned over the feet.
- Do not flex the torso forward or round the back.
- Continue extending the hips and knees to reach the starting position.
- At the end of the set, step forward toward the rack.
- Squat down until the bar rests on the supports.

Upward Movement Phase: Two Spotters